

FORT YUKON, AK, USA

WMO: 701940

Lat: 66.567N

Lon: 145.267W

Elev: 445

StdP: 14.46

Time Zone: -9.00 (NAK)

Period: 82-19

WBAN: 26413

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -54.0	(c) -49.1	(d) -63.0	(e) 0.1	(f) -55.3	(g) -56.9	(h) 0.2	(i) -19.3	(j) 18.9	(k) -2.8	(l) 16.8	(m) -3.8	(n) 0.6	(o) 40	(p) 0.704

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	79.8	59.8	76.8	58.8	73.4	57.6	62.2	76.4	60.3	74.3	58.9	72.1	5.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
56.7	69.6	66.0	54.2	63.6	64.3	52.3	59.3	62.8	28.1	75.9	26.8	74.5	25.7	72.1	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 17.5	(b) 15.2	(c) 12.8	(e) -54.2	(f) 85.0	(g) 5.6	(h) 3.8	(i) -58.2	(j) 87.8	(k) -61.5	(l) 90.0	(m) -64.7	(n) 92.1	(o) -68.8	(p) 94.8		
(4) 17.5	(5) 15.2	(6) 12.8	(7) N/A	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.0	-14.2	-10.4	2.6	24.8	46.4	59.9	62.5	54.8	42.0	19.8	-4.7	-10.1	
	DBStd	31.00	20.99	16.45	12.43	12.03	9.07	6.01	5.20	6.47	7.85	12.25	15.31	16.28	
	HDD50	10815	1991	1690	1469	757	175	6	1	27	255	938	1642	1864	
	HDD65	15397	2456	2110	1934	1207	576	170	109	320	691	1403	2092	2329	
	CDD50	946	0	0	0	0	64	304	387	177	14	0	0	0	
	CDD65	53	0	0	0	0	0	18	30	5	0	0	0	0	
	CDH74	600	0	0	0	0	20	201	305	60	0	0	0	14	
	CDH80	87	0	0	0	0	1	21	47	8	0	0	0	10	
Wind	WSAvg	4.7	3.0	3.3	5.4	6.7	6.6	5.4	5.2	4.8	5.1	4.7	3.3	3.1	
Precipitation	PrecAvg	15.6	1.0	0.8	0.7	0.5	0.9	1.6	2.2	2.4	1.9	1.4	1.1	1.1	
	PrecMax	25.0	2.7	1.9	2.7	1.5	2.2	3.7	4.3	4.0	4.3	3.4	3.1	2.7	
	PrecMin	11.1	0.2	0.0	0.2	0.1	0.3	0.4	1.0	0.3	0.4	0.4	0.1	0.3	
	PrecStd	3.4	0.7	0.4	0.5	0.3	0.4	0.7	0.8	0.7	1.0	0.7	0.7	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	30.4	39.3	57.2	77.0	82.8	84.7	80.5	66.4	47.1	32.2	33.4	
		MCWB	33.3	28.3	32.2	44.0	57.2	59.5	61.9	62.9	52.3	41.0	22.1	32.7	
	2%	DB	36.5	24.2	33.6	52.2	71.9	79.2	80.9	74.1	62.9	40.8	25.0	19.4	
		MCWB	24.5	22.3	28.6	40.6	52.4	57.9	61.2	57.2	52.0	36.2	17.1	18.5	
	5%	DB	18.4	17.9	27.5	47.5	66.5	75.8	77.5	70.2	58.9	36.9	19.3	14.0	
		MCWB	14.8	16.8	22.8	37.9	49.5	57.5	60.7	56.3	49.0	34.0	17.0	13.2	
	10%	DB	9.9	11.5	21.9	43.3	63.2	72.9	74.4	66.2	54.9	34.1	15.9	9.9	
		MCWB	9.0	10.2	18.7	34.9	48.1	56.9	58.7	54.6	46.4	31.7	13.2	9.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.5	30.1	32.2	43.9	58.7	63.3	65.5	63.0	54.6	39.5	28.9	28.2
			MCDB	52.8	32.4	39.0	56.1	78.5	78.0	78.0	75.4	63.1	42.5	43.3	30.0
2%		WB	32.7	23.8	27.8	40.4	52.2	61.0	63.2	59.5	51.0	35.3	18.0	20.3	
		MCDB	50.2	25.9	33.4	50.5	67.0	76.0	76.9	70.9	58.3	37.7	20.1	21.5	
5%		WB	29.2	16.6	22.1	36.9	49.6	59.7	61.3	56.8	48.6	32.8	13.5	14.5	
		MCDB	44.0	17.8	25.6	45.4	62.4	74.5	75.7	67.1	54.8	34.6	15.2	15.3	
10%		WB	17.4	9.3	17.8	33.3	47.5	58.4	59.6	55.3	46.1	31.0	9.6	9.9	
		MCDB	21.2	10.3	20.5	40.3	60.3	72.4	72.7	64.1	52.5	32.7	11.5	10.7	
Mean Daily Temperature Range	5% DB	MDBR	14.1	17.4	22.8	21.8	21.3	19.6	18.1	17.5	17.1	12.1	13.7	15.1	
		MCDBR	17.9	19.0	26.3	23.8	26.1	23.8	24.0	22.8	23.2	13.1	15.0	24.7	
		MCWBR	12.1	18.2	23.5	15.7	13.9	9.8	11.0	12.7	16.1	10.0	16.6	17.6	
	5% WB	MCDBR	13.0	19.8	27.7	23.8	24.2	22.7	23.0	20.0	19.8	11.1	17.8	17.7	
		MCWBR	8.0	18.3	23.2	15.3	13.1	9.6	10.7	11.8	14.3	9.7	15.7	17.4	
Clear-Sky Solar Irradiance	taub	0.165	0.231	0.252	0.328	0.339	0.362	0.381	0.368	0.316	0.250	0.163	N/A		
	taud	1.925	2.192	2.271	2.148	2.424	2.385	2.355	2.396	2.510	2.402	1.955	N/A		
	Ebn at Noon	82	206	263	261	273	268	257	248	239	201	78	0		
	Edh at Noon	5	15	24	36	30	32	33	28	20	12	4	0		
All-Sky Solar Radiation	RadAvg	28	228	795	1486	1822	1968	1679	1217	736	254	50	0		
	RadStd	3	15	39	73	128	143	109	102	78	37	6	0		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page